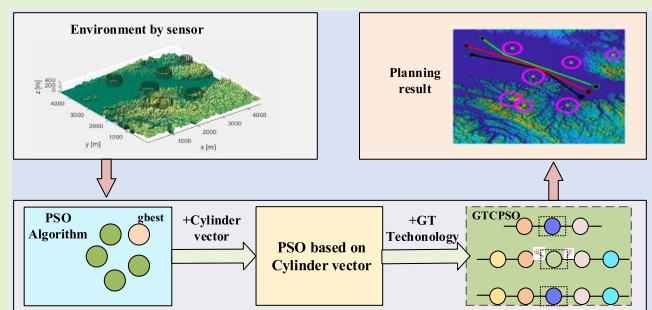


Cooperative Path Planning of Multiple Unmanned Aerial Vehicles Using Cylinder Vector Particle Swarm Optimization With Gene Targeting

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Abstract—Cooperative route planning is crucial to ensure safe and effective flight of unmanned aerial vehicle (UAV) formation. Complicated scenarios, however, put forward new challenges for path planning of multiple UAVs; therefore, a novel cylindrical coordinate particle swarm optimization (PSO) planner with gene targeting (GTCPSO) is presented to obtain feasible and accurate flight routes for multiple UAVs. In the GTCPSO planner, to satisfy UAV dynamic characteristics, the cylindrical coordinate is exploited to represent the flight route; the PSO algorithm with cylindrical coordinates can quickly find a feasible trajectory in a complex environment. Meanwhile, by incorporating gene targeting (GT) operation into improved cylindrical coordinate PSO, the bottleneck waypoint in the global best trajectory is found by waypoint detection, and then the detected bottleneck waypoint is modified by the adjustment operation based on two-phase the updating strategy. A series of experiments based on real map examples on multi-sensors are designed to test the search capability of this GTCPSO planner. The simulation results indicate that the proposed planner GTCPSO has the potential to handle multi-UAV cooperative planning problems under complex constraints.

Index Terms—Cylinder vector, gene targeting, multiple unmanned aerial vehicles (UAVs), multisensors, particle swarm optimization (PSO), path planning.



I. INTRODUCTION

UNMANNED aerial vehicles (UAVs) have gained wide attention because of their unique capability and robust-

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ness [1], [2]. In the past decades, UAVs have been widely applied in reconnaissance and surveillance, express transportation, and rescue missions. Flight route planning is a key technology of UAV navigation system, and its aim is to quickly find multiple flight waypoints from the starting position to the goal under mission constraints. Aiming to enhance the efficiency of completing complicated tasks, UAVs appear in the form of clusters in most task scenarios. Compared with a single UAV, the cooperative control of multiple homogeneous or heterogeneous UAVs collaboratively performs tasks by sharing information to improve work efficiency [3]. There have been relevant studies on multi-UAV formation control. Classical UAV formation control methods can be divided into leader-follower method, behavior-based method, virtual rigid body method, etc. UAV formation control is to solve the possible conflicts and interactions among group members under various constraints to maintain the ideal shape when performing robot tasks [4].

The traditional path planning algorithms based on the map mainly include the A* algorithm [5], the rapidly exploring random trees (RRT) algorithm [6], the artificial potential

field (APF) algorithm [7], etc. These traditional planners are, however, difficult to deal with these complex environments. The tasks of route planning for controlling multiple UAVs may be framed as complex optimization problems under numerous constraints and are developed by evolutionary computation (EC) algorithms. Compared with traditional algorithms, various metaheuristic algorithms, including genetic algorithm (GA) [8], [9], differential evolution (DE) [10], [11], artificial bee colony (ABC) [12], [13], particle swarm optimization (PSO) [14], [15] and so on [16], [17], [18], and [19], can seek a better route under complex environment. Because of fast convergence, high optimization accuracy, high efficiency, and good stability, metaheuristic algorithms are widely adopted to handle complex optimization issues and challenges.

PSO is designed to handle many optimization problems [20]. It depends on information sharing among the individuals in a population, and the population becomes reasonably ordered in the search space. After the initialization of the PSO algorithm, each particle is evaluated by objective function, and then continues to seek for better solution. The population can find the optimal solution through the iteration operation. All individuals in the population update their velocity and position according to the personal best position and the global best position of the population as follows:

$$\begin{cases} v_i(t+1) = \omega v_i(t) + c_1 \text{rand}_1(x_{i,pbest}(t) - x_i(t)) \\ \quad + c_2 \text{rand}_2(x_{gbest}(t) - x_i(t)) \\ x_i(t+1) = x_i(t) + v_i(t+1) \end{cases} \quad (1)$$

where $v_i(t)$ represents the velocity of the i th particle at time t , $x_i(t)$ is the corresponding position of the i th particle, c_1 and c_2 are the cognitive and social coefficients, rand_1 and rand_2 are samples uniformly distributed in the range of $[0, 1]$, and ω is the inertia weight.

Although PSO is robust and simple for path planning, it also has some shortcomings, e.g., a premature and long optimization time under a complex environment. With the increasing complexity of the task, many researchers have put more attention to enhancing the search efficiency. A route planning approach of multiple UAVs is presented by introducing multimode collaboration [21]. Shao et al. [22] presented adaptive linear-varying acceleration coefficients and mutation strategies to find the optimal path for UAV formation. Xu et al. [23] designed a modified PSO algorithm by introducing dynamic multiswarm and comprehensive learning to search UAV flight paths. Phung and Ha [24] introduced a spherical vector-based PSO to seek feasible routes in complicated environments.

Gene targeting (GT) is a genetic technique that uses homologous recombination to alter an organism's endogenous gene. With gene recombination, target genes are recombined with homologous DNA on the chromosomes of immune cells. Target genes are integrated into a certain site in the genome of immune cells to improve or change the genotype of immune modulatory cells. GT technique is often referred to as gene knockout. It involves that the structure of this gene is known, but its function is unknown, and through designing experiments at the molecular level to remove that gene or replace it with a similar gene. In 2007, the Nobel Prize was awarded to

Capecchi, Smith, and Evans for stem cell research, particularly in the development of GT technology. GT technology has also been applied to the medicine and cosmetics fields. Wang et al. [25] provided a simple and effective DE (GTDE) algorithm by using GT to deal with large-scale optimization problems.

Inspired by GT technology, an improved PSO path planning method by combining cylinder vector and GT is presented in this article. First, GT constructs a gene detector, which is adopted to find the bottleneck waypoint in the global best path. Then, the bottleneck waypoint is replaced by the homologous targeting vector to improve the quality of the global best route. In the process of replacing the waypoint, considering the search characteristics of PSO, the homologous targeting vector is generated by a two-stage search strategy to enhance search efficiency. The contribution of this article mainly includes the following two aspects.

- 1) An improved PSO planner based on cylindrical coordinates and GT is developed to seek flight routes for multiple UAVs. By identifying the bottleneck waypoint in the current global best route and updating the poor points, a more accurate global best route is generated with a high success rate.
- 2) A two-phase search strategy by considering PSO evolution process is designed to obtain new targeting vectors. Through two-stage search strategy, the flight path generated by new planner can satisfy the requirements of multiconstraint and multiobjective for path planning.

The remainder of this article is organized as follows. Section II formulates mathematical model of path problem, Section III describes the proposed algorithm in detail. Section IV analyzes the experiment results, and Section V summarizes the conclusion and looks forward to future work.

II. PROBLEM DESCRIPTION

A. Flying Space and Path Representation

The flying space is the area where the UAV implements the flight task. In 3-D space, we denote $M(x, y, z)$ to be a waypoint in flight space, and the polar coordinate of the projection W_P of waypoint M is (r, θ) . r represents the length of OM , θ represents the angle from Ox to OM , r is the polar diameter of the point M , θ is the polar angle of the point M ; therefore, (r, θ, z) is denoted as the cylindrical coordinate of waypoint M . The waypoint M is represented by the cylinder coordinate

$$\Omega = \{(r, \theta, z) \mid -r_{\max} \leq r \leq r_{\max}, -\theta_{\max} \leq \theta \leq \theta_{\max}, -z_{\max} \leq z \leq z_{\max}\} \quad (2)$$

where Ω represents the range of the task scene, $-r_{\max}$, $r_{\max}$, $-\theta_{\max}$, $\theta_{\max}$, $-z_{\max}$ and $z_{\max}$ represent the minimum and maximum value of r , θ and z coordinate, respectively.

B. Objective Function

In the flight route planning of multiple UAVs, the path length, the terrain collision, and the radar threats should be considered. Besides that, it is necessary to take the space collision between the multiple UAVs into account, that is, to maintain an appropriate safe distance between all UAVs

during the flight. The objective function is, therefore, given by the following equation:

$$f = a_1 f_L + a_2 f_T + a_3 f_H + a_4 f_S + a_5 f_C \quad (3)$$

where f_L is flight route length cost, f_T is radar threat cost, f_H is height cost, f_S is smoothness cost, f_C is UAV collision cost, and a_1 – a_5 are the weight coefficients.

1) Route Length Cost f_L : Route length cost is an important index that determines the quality of flight trajectory, which determines the time and energy consumption during the flight. A shorter flight path length can reduce energy consumption and ensure safety. The length cost of the flight route can be defined as follows:

$$f_L = \sum_{j=0}^N \left\| \overrightarrow{W_j W_{j+1}} \right\| \quad (4)$$

where N represents the number of the waypoints except the start point and the goal point; W_j represents the coordinates of j th waypoint in the route. When $j = 0$, W_j represents the starting point. When $j = N$, W_{N+1} represents the goal. $\|W_j W_{j+1}\|$ is the Euclidean distance between W_j and W_{j+1} .

2) Threat Cost f_T : The flight area is usually threatened by the radars and the missiles. In order to improve the survival probability of the UAV, the path should avoid flying into the threat area. The threat cost of $\|W_j W_{j+1}\|$ is calculated as follows [23]:

$$\begin{cases} f_T = \sum_{j=1}^{N-1} \sum_{k=1}^K T_k(\overrightarrow{W_j W_{j+1}}) \\ T_k(\overrightarrow{W_j W_{j+1}}) = \begin{cases} 0, & \text{if } d_k - r_k > s + D \\ r_k + s - d_k, & \text{if } D < d_k - r_k \leq s + D \\ C, & \text{if } d_k - r_k \leq D \end{cases} \end{cases} \quad (5)$$

where K is the number of the threat radars and the missiles; d_k is the distance between the j th segment and the center of the k th radar threat. r_k is the radius of the k th radar threat. D is determined by the UAV size. s is the safety margin. C is the penalty constant to punish the routes into the radar threat area.

3) Height Cost f_H : If the flying height of the UAV is too low, it will increase the probability of collision with the mountain terrain. When the flying height is too high, it will reduce the endurance of the UAV. The height cost is, therefore, calculated by the following equation:

$$\begin{cases} f_H = \sum_{j=1}^N H_j \\ H_j = \begin{cases} \left| h_j - \frac{(h_{\max} + h_{\min})}{2} \right|, & \text{if } h_{\min} \leq h_j \leq h_{\max} \\ C, & \text{otherwise} \end{cases} \end{cases} \quad (6)$$

where H_j is the flight height cost associated with waypoint W_j , $h_{\max}$ and $h_{\min}$ are the minimum and maximum height values, respectively. C is the penalty constant.

4) Smoothness Cost f_S : Because of the dynamic constraints of the UAV, it is impossible to make sharp turn, so the algorithm is required to generate a smooth path. This can be implemented by restraining the climbing angle and turning angle.

The climbing angle is used to quantify the maneuvering performance of the UAV as follows [3]:

$$\alpha_j = \arctan \left(\frac{z_{j+1} - z_j}{\left\| \overrightarrow{W'_j W'_{j+1}} \right\|} \right) \quad (7)$$

where z_j and z_{j+1} are z coordinate of the j th waypoint and $(j + 1)$ th waypoint. $\|\overrightarrow{W'_j W'_{j+1}}\|$ is the distance between the projections W'_j of the waypoint W_j and W'_{j+1} of the waypoint W_{j+1} .

The turning angle β_j is calculated as follows:

$$\beta_j = \arccos \left(\frac{\overrightarrow{W'_j W'_{j+1}} \cdot \overrightarrow{W'_{j+1} W'_{j+2}}}{\left\| \overrightarrow{W'_j W'_{j+1}} \right\| \cdot \left\| \overrightarrow{W'_{j+1} W'_{j+2}} \right\|} \right) \quad (8)$$

where W'_j , W'_{j+1} , and W'_{j+2} are the projections of the waypoint W_j , W_{j+1} , and W_{j+2} on the horizontal plane Oxy plane.

The smoothness cost can be calculated as follows:

$$f_S = \sum_{j=1}^{N-1} \alpha_j + \sum_{j=1}^{N-2} \beta_j. \quad (9)$$

5) UAV Collision Cost f_C : The distance between multiple UAVs should keep a safe threshold value L_{safe} to avoid collision during the flight. When the distance L_{ij} between two UAVs is less than the safety clearance L_{safe} , there will exist a collision risk between them. The collision cost between multiple UAVs is, therefore, calculated as follows:

$$\begin{cases} f_C = \sum_{i=1}^N \sum_{j=1}^N C_{ij} \\ C_{ij} = \begin{cases} C, & \text{if } L_{ij} \leq L_{\text{safe}} \\ 0, & \text{otherwise} \end{cases} \end{cases} \quad (10)$$

where L_{ij} is the distance between the waypoint W_i and the waypoint W_j of other UAVs.

III. PROPOSED ROUTE PLANNER GTCP SO

Because the polar coordinate r and θ in the cylindrical coordinates conforms to the dynamic characteristics of the UAV, using cylindrical coordinates to seek the waypoint can find the feasible path more conveniently and quickly. The optimization mechanism of PSO is that all particles learn from the global best solution in the population to achieve the updating operation. The bottleneck waypoint in optimal path will, however, seriously affect the performance of the whole path. By introducing GT technology, the optimal route is, therefore, improved by bottleneck waypoint detection and the targeting vector reconstruction. The specific process is described in Sections III-A–III-B.

A. Cylinder Vector-Based PSO for UAV Path Planning

The cylindrical coordinate is adopted to represent the flight path of UAV, which can better satisfy the dynamic characteristics of UAV. Each path in 3-D space is denoted by a cylindrical coordinate replacing the original Cartesian coordinate. We assume the cylindrical coordinate of j th waypoints of particle i be $(r_{ij}, \theta_{ij}, z_{ij})$, then a flight path consisting of N waypoints is defined by a cylindrical coordinate in 3-D space as follows:

$$p_i = (r_{i1}, \theta_{i1}, z_{i1}, r_{i2}, \theta_{i2}, z_{i2}, \dots, r_{iN}, \theta_{iN}, z_{iN}). \quad (11)$$

Define p_i the position of a particle, then the velocity corresponding to this cylindrical vector is represented by an incremental vector

$$\Delta p_i = (\Delta r_{i1}, \Delta \theta_{i1}, \Delta z_{i1}, \Delta r_{i2}, \Delta \theta_{i2}, \Delta z_{i2}, \dots, \Delta r_{iN}, \Delta \theta_{iN}, \Delta z_{iN}). \quad (12)$$

The updated equations for cylindrical coordinate particle swarm optimization (PSO) planner with gene targeting are, therefore, shown as follows:

$$\begin{cases} p_{ij}(t+1) = \omega p_{ij}(t) + c_1 \text{rand}_1(p_{bij}(t) - p_{ij}(t)) \\ \quad + c_2 \text{rand}_2(g_j(t) - p_{ij}(t)) \\ p_{ij}(t+1) = p_{ij}(t) + p_{ij}(t+1) \end{cases} \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, N) \quad (13)$$

where $p_{bi}(t)$ and $g(t)$ are personal best position of particle i and the global best position of the swarm, respectively. rand_1 and rand_2 are samples uniformly distributed in the range of $[0, 1]$.

In order to evaluate the quality of the route, the cylindrical vector needs to be converted into a Cartesian coordinate by the following equation:

$$\begin{cases} x_{ij} = x_{i,j-1} + r_{ij} \cos \theta_{ij} \\ y_{ij} = y_{i,j-1} + r_{ij} \sin \theta_{ij} \\ z_{ij} = z_{i,j-1} + \Delta z_{ij} \end{cases} \quad (14)$$

where $(x_{i,j-1}, y_{i,j-1}, z_{i,j-1})$ and $(x_{i,j}, y_{i,j}, z_{i,j})$ are $(j-1)$ th and j th waypoints of particle i in Cartesian coordinate system. $(r_{ij}, \theta_{ij}, z_{ij})$ is j th waypoints of particle i in cylindrical coordinate system.

B. Gene Targeting

In PSO, all the particles depend on the guidance of global best solution, so global best solution has an impact on the solution accuracy of PSO. GT technology is introduced into the global best route, which can efficiently improve the quality of the global best path.

1) Bottleneck Waypoint Detection: When GT technology is employed to deal with the path planning problem, first, the bottleneck waypoint will directly influence the current global best path and can be regarded as pathogenic genes. According to the objective function for multiple UAVs, a simple and effective detection function is established to detect bottleneck waypoints of the global optimal path. We exploit the increment of the objective function value of each waypoint in the current

global best path as a detection function. The detection function ΔF_{gi} of i th waypoint in current global best path is given by the following equation:

$$\Delta F_{gi} = F_{gi} - F_{g(i-1)}, \quad (i = 2, 3, \dots, N) \quad (15)$$

where ΔF_{gi} is the increment of i th waypoint fitness value in the current global best path, F_{gi} and $F_{g(i-1)}$ are objective function values of i th waypoint and $(i-1)$ th waypoint in the current global best path, respectively.

The objective function F_{gi} of i waypoint in current global best path can be given as follows:

$$F_{gi} = a_1 f_{gLi} + a_2 f_{gTi} + a_3 f_{gHi} + a_4 f_{gSi} + a_5 f_{gCi}, \quad (i = 1, 2, \dots, N) \quad (16)$$

where f_{gLi} is the flight route length cost of i th waypoint in the current global best path, f_{gTi} is radar threat cost containing i th waypoint in the current global best path, f_{gHi} is height cost containing i th waypoint in the current global best path, f_{gSi} is smoothness cost containing i th waypoint in the current global best path and f_{gCi} is UAV collision cost containing i th waypoint in the current global best path, a_1 – a_5 are the corresponding weight coefficients.

By means of (16), the detection function ΔF_{gi} value of each waypoint is calculated by the bottleneck waypoint detection. Specifically, for each waypoint of the global best route, probability p is calculated to determine the bottleneck waypoint of the global best route. $p(i)$ can be given by the following equation:

$$p(i) = \frac{\Delta F_{gi}}{\sum_{i=1}^N \Delta F_{gi}}. \quad (17)$$

The waypoint with the maximum p value will be selected as the bottleneck waypoint. Otherwise, the waypoint is treated as an ordinary waypoint. The process for bottleneck waypoint detection is shown in Algorithm 1.

Algorithm 1 Detect the Bottleneck Waypoint

1. **For** $i = 1$ to N
 2. Calculate the detection function value ΔF_{gi} by Eq.(15)
 3. **End**
 4. Calculate $p(i)$ of the i th waypoint in the global best route by Eq.(17)
 5. The waypoint with the maximum p value is selected as the bottleneck
 6. waypoint
-

2) Generate New Targeting Vectors: DNA recombination technique is used to construct new targeting waypoint. Here, only the bottleneck waypoint of the global best route is reconstructed based on GT technology, which will be much easier and more efficient. In order to increase the diversity of the algorithm, the Monte Carlo method is adopted to select some particles as candidate particles from the current particles according to a given sample probability. Meanwhile, we consider that the PSO algorithm is likely close to the global optimal solution in the evolution process. A two-stage search strategy is, therefore, designed to generate new targeting waypoint of the global best solution.

First, Monte Carlo method is employed to obtain the candidate targeting vector with the probability p_i . Then, considering PSO algorithm may lead a lot of particles converge to the global best solution after a certain stage of the evolution process, the perturbation is added into the routes randomly sampled from the swarm to help the swarm jump out of the local optimal.

Algorithm 2 Generate New Targeting Vectors

```

1. If  $g < g_1$ 
2.   For  $i = 1$  to  $m$ 
3.      $P_i = \text{Gaussian}(0.2, 0.2)$ ;
4.     If  $\text{rand} < P_i$ 
5.       The  $i$ th route is selected as the candidate targeting vector
6.     End if
7.   End for
8. End If
9. Else
10.  For  $i = 1$  to  $m$ 
11.     $P_i = \text{Gaussian}(0.2, 0.2)$ 
12.    If  $\text{rand} < P_i$ 
13.      The  $i$ th particle is selected and generate the candidate
14.      targeting vector using (18)
15.    End if
16.  End for
17. End if

```

Let g represent the current number of iterations, and g_1 is a fixed number of iterations. If $g < g_1$, a Monte Carlo-based approach is used here to obtain a homologous targeting vector with a probability of p_i . While the current generation $g > g_1$, the candidate route is obtained by the perturbation operator of current route, which are given as follows:

$$\begin{cases} r_i^{\text{new}} = r_i + 10 * (\text{rand} - 0.5) * \text{Vel}_{\text{Max}} \cdot r \\ \theta_i^{\text{new}} = \theta_i + 10 * (\text{rand} - 0.5) * \text{Vel}_{\text{Max}} \cdot \theta \\ z_i^{\text{new}} = z_i + 10 * (\text{rand} - 0.5) * \text{Vel}_{\text{Max}} \cdot z \end{cases} \quad (18)$$

where rand is a random number belong to $(0, 1)$, $(r_i^{\text{new}}, \theta_i^{\text{new}}, z_i^{\text{new}})$ are the new targeting vectors, (r_i, θ_i, z_i) is current waypoint, $\text{Vel}_{\text{Max}} \cdot r$, $\text{Vel}_{\text{Max}} \cdot \theta$, and $\text{Vel}_{\text{Max}} \cdot z$ are lower and upper bounds of the velocity.

3) New Targeting Waypoint Addition: After constructing the targeting vectors, we obtain m_1 new candidate routes. The number of these candidate routes is assumed to be m_1 . The bottleneck waypoint of the global best path P_g will be modified by the corresponding waypoint of these candidate routes, while the other waypoints in the global best path will be preserved. Mutant embryonic stem cells, i.e., newly generated paths, are recorded as P'_i . The whole process of inserting homologous target vectors is shown in Algorithm 3.

The above three steps constitute a complete search process based on GT technology. GT technology is used to generate new paths P' . Compare the fitness values of these new routes P' and the original global best path P_g before GT, if the fitness value of the newly generated route is superior to the original global best path P_g before GT, this newly generated route with the smaller fitness value will replace the original global

Algorithm 3 New Targeting Waypoint Addition

```

1. For  $i_1 = 1 : m_1$ 
2.   For  $j = 1$  to  $N$ 
3.     If the  $j$ th waypoint is the bottleneck waypoint
4.        $P'_{i_1j} = P_{i_1j}$ 
5.     Else
6.        $P'_{i_1j} = P_{gj}$ 
7.     End If
8.   End For
9. End for
10. Obtain  $m_1$  new paths

```

best path as the new global best route. In this method, the bottleneck waypoint in the global optimal path is checked by the waypoint detection function, and new targeting vectors are constructed by the two-stage strategy, which provides more promising examples for the evolution of the algorithm and enhances the potential in exploiting the flight space.

4) Complete GTCP SO Algorithm: In the GTCP SO algorithm, first, randomly initialize the routes with cylindrical coordinates and then compute the fitness value of each path at steps 1–4. GTCP SO records the global optimal route based on fitness. Second, m routes are updated by (13) at steps 6–10. For the global best route, GTCP SO employs Algorithm 1 to detect the bottleneck waypoint by the waypoint detection function at step 11. New targeting vectors are generated by Algorithm 2 at step 12. The targeting vectors can easily be incorporated into the global best path by Algorithm 3 at step 13, and after that, m_1 new routes are obtained. Compare the new routes and original global best route, and select the smaller fitness value as new global best route at steps 14–18. After the whole optimization process is iterated for T_{max} times, the best solution after T_{max} iterations is output at step 20. The optimal path is constantly corrected and self-updated based on GT technology in each iteration, so the whole population can gradually converge to a better global optimal by learning from the optimal path.

C. Implementation of GTCP SO for Multiple UAVs

1) Range of r and θ in Cylindrical Coordinates: The range of r is given by the following equation:

$$0 \leq r \leq \frac{2D_{ST}}{N} \quad (19)$$

where D_{ST} is the distance between the start point and endpoint of the flight route. N is the number of the waypoint.

The range of θ is given as follows:

$$\theta_{\min} \leq \theta \leq \theta_{\max} \quad (20)$$

where $\theta_{\min}$ and $\theta_{\max}$ represent the maximum and minimum of the angle, respectively,

$$\begin{cases} \theta_{\min} = \arctan \frac{y_{W_G} - y_{W_S}}{x_{W_G} - x_{W_S}} - \theta_r \\ \theta_{\max} = \arctan \frac{y_{W_G} - y_{W_S}}{x_{W_G} - x_{W_S}} + \theta_r \end{cases} \quad (21)$$

Algorithm 4 GTCPSO

1. $\omega = 0.73$, $c_1 = c_2 = 1.5$, $g = 0$;
2. Randomly generate m routes based on Cylindrical coordinates;
3. Calculate the fitness value of m routes using (2);
4. Find the global best route P_g ;
5. **While** $g < T_{max}$
6. **For** $i = 1$ to m
7. Update the m routes by (13)
8. Calculate the fitness value of m routes using (2);
9. Find the global best route P_g ;
10. **End for**
11. Detect the bottleneck waypoint of the global best route through Algorithm 1;
12. Obtain the targeting vectors by Algorithm 2;
13. Put the targeting vectors to the global best path and obtain m_1 new route by Algorithm 3;
14. **For** $i_1 = 1$ to m_1
15. **If** $f(P'_{i_1}) < f(P_g)$
16. $P_g = P'_{i_1}$
17. **End If**
18. **End For**
19. $g = g + 1$;
20. **End while**

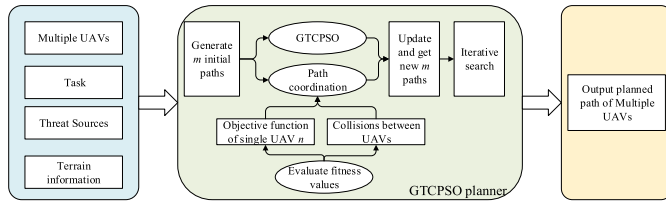


Fig. 1. Framework of multiple UAVs path planning based on GTCPSO.

where (x_{WS}, y_{WS}) and (x_{WG}, y_{WG}) are the x and y coordinates of the starting and the goal, respectively. θ_r is a fixed angle.

To ensure that UAV can fly safely and quickly to the destination, θ_r is determined by the following steps. First, we establish an angle selection set $S_{\theta_r} = [(\pi/8), (\pi/6), (\pi/4), (\pi/2)]$. Then, the candidate angles in the set S_{θ_r} are evaluated by a parallel feedback method [14]. Finally, the path obtained from the parallel feedback method with the smallest fitness value is selected as global best route of the initial path, and the corresponding angle value in S_{θ_r} is chosen as θ_r .

2) Implementation of GTCPSO for Multiple UAVs: The framework of multiple UAVs cooperative flight path planning based on GTCPSO is displayed in Fig. 1.

The implementation procedures of GTCPSO for path planning of multiple UAVs are described as follows.

Step 1: Input the 3-D environment information, such as planning space, the obstacle center, the obstacle size, and the threat sources.

Step 2: For all N UAVs, set the population size to m , the maximum number of iterations to T_{max} , and the current number of iterations to $t = 0$, initial the weight ω and acceleration coefficients c_1, c_2 .

Step 3: Initialize m paths represented by cylindrical coordinates. Convert cylindrical coordinates into the corresponding Cartesian coordinates by the mapping function defined in (14).

Step 4: Then, evaluate each path according to (3) and select the global best route according to the fitness value.

Step 5: GT technology is adopted to detect the bottleneck point of the global optimal path, and generate some candidate paths. Compare the fitness values of new candidate paths and the original global best route, determine the global best route.

Step 6: Update each path according to (13).

Step 7: If the algorithm reaches the maximum number of iterations, it goes to step 8; otherwise, it turns to step 4.

Step 8: Output the current global best route as the result when the loop ends.

IV. EXPERIMENTAL RESULTS AND COMPARISONS

In order to verify the performance of GTCPSO in path planning for UAVs, the 3-D mountain terrain is established by the digital elevation map [26] and the simulation experiment are carried out. The simulation is implemented in 3.00 GHz CPU, 32.0 GB of RAM and Windows 10 system.

A. Parameter Setting

The proposed GTCPSO is compared with standard PSO [27] and some state-of-the-art PSO variant algorithms, including PSO-LDIW [28], θ -PSO [29], APSO [30], AWPSSO [31], PSO-TVAC [32], SPSO [24], GA [27], TLBO [33] and FFA [34]. PSO-LDIW is the PSO with linear weight decline. θ -PSO is the PSO based on angle change. APSO is the PSO with adaptive adjustment parameters according to the population state. AWPSSO is a PSO variant based on an S-type function weighting strategy, and PSO-TVAC is a self-organizing stratified PSO with time-varying acceleration coefficients. SPSO is a modified PSO based on spherical coordinates. The relevant parameters of different PSO algorithms are shown in Table I. Each algorithm is independently run 25 times. The population size of all comparison algorithms is set to $m = 100$ and the maximum iteration is $T_{max} = 100$, to be fair.

B. Experiment for Multiple UAVs

In order to evaluate the performance of the GTCPSO algorithm, in this section, GTCPSO and other PSOs are used to plan routes for multiple UAVs. The safe distance between UAVs must be considered as crucial indicator. Two experiments are carried out in scenarios 1 and 2. In scenario 1, three UAVs (UAV 1, UAV 2, UAV 3) fly from different starting points (300, 1000, 150), (50, 850, 150), and (700, 1150, 150) to the same target point (850, 50, 150). We design scenario 2, including five UAVs that fly from starting points (50, 300, 100), (50, 500, 100), (50, 700, 100), (50, 100, 100), and (50, 900, 100) to target points (950, 400, 100), (950, 450, 100), (950, 500, 100), (950, 350, 100), and (950, 550, 100).

Fig. 2–Fig. 4 show the flight path diagram, fitness value comparison and convergence curve comparison of 3 UAVs in scene 1. Figs. 3–5 display the flight route, fitness value comparison and convergence curve comparison of 5 UAVs in scene 2. Tables II and III show the statistic results of multiple

TABLE I

PARAMETER SETTINGS IN VARIOUS COMPARISON ALGORITHMS

Algorithm	Parameters	Reference
PSO	$\omega = 0.73$, $c_1 = c_2 = 1.5$	[27]
PSO-LDIW	$\omega: 0.9-0.4$, $c_1 = c_2 = 2$	[28]
θ -PSO	$\omega(t) = \omega(t-1) \times damp$, $\omega(0) = 1$, $damp = 0.98$, $c_1 = c_2 = 1.5$	[29]
APSO	Automatically chosen	[30]
AWPSO	Automatically chosen	[31]
PSO-TVAC	$\omega: 0.9-0.4$, $c_1: 2.5-0.5$, $c_2: 0.5-2.5$	[32]
SPSO	$\omega(t) = \omega(t-1) \times damp$, $\omega(0) = 1$, $damp = 0.98$, $c_1 = c_2 = 1.5$	[24]
GA	Crossover rate: 0.8; Mutation rate: 0.1	[27]
TLBO	$T_F = \text{round}[1 + \text{rand}(0,1)]$	[33]
FFA	$\beta_0 = 1, \alpha = 1, \gamma = 0.1$	[34]
GTCP SO	$\omega = 0.73$, $c_1 = c_2 = 1.5$	Proposed

TABLE II

PERFORMANCE COMPARISON OF VARIOUS ALGORITHMS IN SCENARIO 1

	Algorithm	Best	Median	Mean	Worst	Std.
UAV1	PSO	6642	8846	9785	14677	3125
	PSO-LDIW	6333	11148	10761	13326	2159
	θ -PSO	6215	8204	8169	9357	899
	APSO	6515	8246	9022	11361	1491
	AWPSO	6212	9934	9032	10235	1402
	PSO-TVAC	5801	7114	7300	9561	1232
	SPSO	7434	8193	8204	8874	392
	GA	6167	9689	9214	11758	2146
	TLBO	7619	9047	9096	10673	1186
	FFA	5784	8430	8019	9433	1358
UAV2	GTCP SO	5588	5847	5893	6650	205
	PSO	7435	9149	9148	10980	1011
	PSO-LDIW	6402	10085	9570	11358	1486
	θ -PSO	6268	9241	8733	10381	1303
	APSO	6172	7840	7549	9131	1092
	AWPSO	6333	7210	7115	10017	1224
	PSO-TVAC	6106	8684	8450	10225	1377
	SPSO	6548	7566	7818	9335	930
	GA	7346	8606	8743	10196	991
	TLBO	6917	7651	7654	8396	534
UAV3	FFA	7512	9073	8837	9689	815
	GTCP SO	5818	6012	6014	6077	39
	PSO	6911	8954	9344	12710	2191
	PSO-LDIW	6916	9252	9919	12782	2377
	θ -PSO	6682	8105	7698	8746	675
	APSO	8051	9346	9304	10585	738
	AWPSO	6459	7805	7647	11418	1670
	PSO-TVAC	6217	7062	7639	11887	1876
	SPSO	7503	8103	8085	8492	255
	GA	6315	6671	7304	8503	970
UAV4	TLBO	6694	10157	9493	10962	1495
	FFA	6001	6125	6459	7960	686
	GTCP SO	5855	6033	6064	6408	96

TABLE III

PERFORMANCE COMPARISON OF VARIOUS ALGORITHMS IN SCENARIO 2

	Algorithm	Best	Median	Mean	Worst	Std.
UAV1	PSO	7194	9872	10322	13012	1780
	PSO-LDIW	9001	9228	9873	10265	1088
	θ -PSO	5681	8092	7705	8161	795
	APSO	6400	9369	9206	11192	1461
	AWPSO	6595	8470	8681	10916	1476
	PSO-TVAC	5212	8102	8108	10687	1478
	SPSO	7162	8260	8269	8879	518
	GA	6747	8042	8124	9666	1206
	TLBO	6816	7665	8607	12806	1978
	FFA	6119	7405	7408	8703	1152
UAV2	GTCP SO	4735	4747	4768	4853	41
	PSO	5165	7876	8535	12631	2293
	PSO-LDIW	5137	6249	6408	7891	869
	θ -PSO	6492	7300	7607	9120	950
	APSO	5669	6755	6787	8278	856
	AWPSO	5059	6788	6421	7601	994
	PSO-TVAC	4690	4857	5226	6958	745
	SPSO	6665	8334	8421	10195	1311
	GA	7748	7942	8094	8744	347
	TLBO	8542	9620	9612	10704	651
UAV3	FFA	7147	7230	7235	7504	81
	GTCP SO	4684	4724	4771	4899	78
	PSO	6548	11062	11214	14721	2274
	PSO-LDIW	8358	8686	8686	9276	261
	θ -PSO	6353	8080	7926	9918	1078
	APSO	6170	8333	8552	10801	1814
	AWPSO	6026	6955	7198	8989	894
	PSO-TVAC	5842	6483	6816	8142	886
	SPSO	6961	7359	7606	9312	738
	GA	6930	7361	7526	8451	533
UAV4	TLBO	6799	9300	9372	11227	1355
	FFA	6002	6198	6192	6371	134
	GTCP SO	4880	4935	4968	5209	102
	PSO	5858	8396	8891	12098	1987
	PSO-LDIW	6332	8995	8995	11533	1473
	θ -PSO	5197	7651	7651	10508	2088
	APSO	5566	6915	6924	8729	893
	AWPSO	5576	6850	6737	8210	812
	PSO-TVAC	5018	7083	7107	9347	1251
	SPSO	5985	7112	6955	7427	425
UAV5	GA	5682	6437	7260	10486	1711
	TLBO	5798	7588	7938	10453	1819
	FFA	5395	5452	5468	5574	58
	GTCP SO	4994	5022	5032	5189	29
	PSO	8024	10440	10136	11810	1345
	PSO-LDIW	6214	6634	7486	11689	1700
	θ -PSO	5909	9389	8583	10686	1682
	APSO	7304	8943	8662	9378	711
	AWPSO	6061	6587	6836	8709	845
	PSO-TVAC	5490	7422	7530	9421	1310
UAV6	SPSO	6435	7042	7242	8310	710
	GA	6070	6322	6412	6933	284
	TLBO	5976	7325	7383	9208	971
	FFA	5653	5673	5685	5842	61
	GTCP SO	5270	5302	5332	5448	56

UAVs' cooperative planning in two scenarios, including the mean fitness value, the median, the worst fitness cost value, and the standard deviation in 25 runs of each algorithm.

For scenario 1, the performance of the PSO algorithm and its variant algorithms are analyzed by comparing the fitness value. It can be seen from Fig. 3 that the average fitness values of UAV 1, UAV 2, and UAV 3 in scenario 1 are obviously different. From Table II, it is clear that the mean fitness values by adopting each improvement are lower and more stable than that of PSO except PSO-LDIW, while the fitness values

of UAV 1, UAV 2, and UAV 3 in the GTCP SO algorithm maintains the lowest and most stable mean fitness value. The performance of SPSO is superior to the other five algorithms for UAV.

As can be seen from Fig. 2, the planned path of each UAV generated by the GTCP SO algorithm is shorter and smoother than other comparative algorithms. The convergence curve in Fig. 6 indicates that GTCP SO algorithm has strong ability in enhancing the solution accuracy and maintaining faster convergence speed. It can be concluded that the proposed GTCP SO can obtain the best performance and improve the path planning performance of multiple UAVs. For the scenario 2, as shown in Figs. 4–7 and Table III, Table III shows

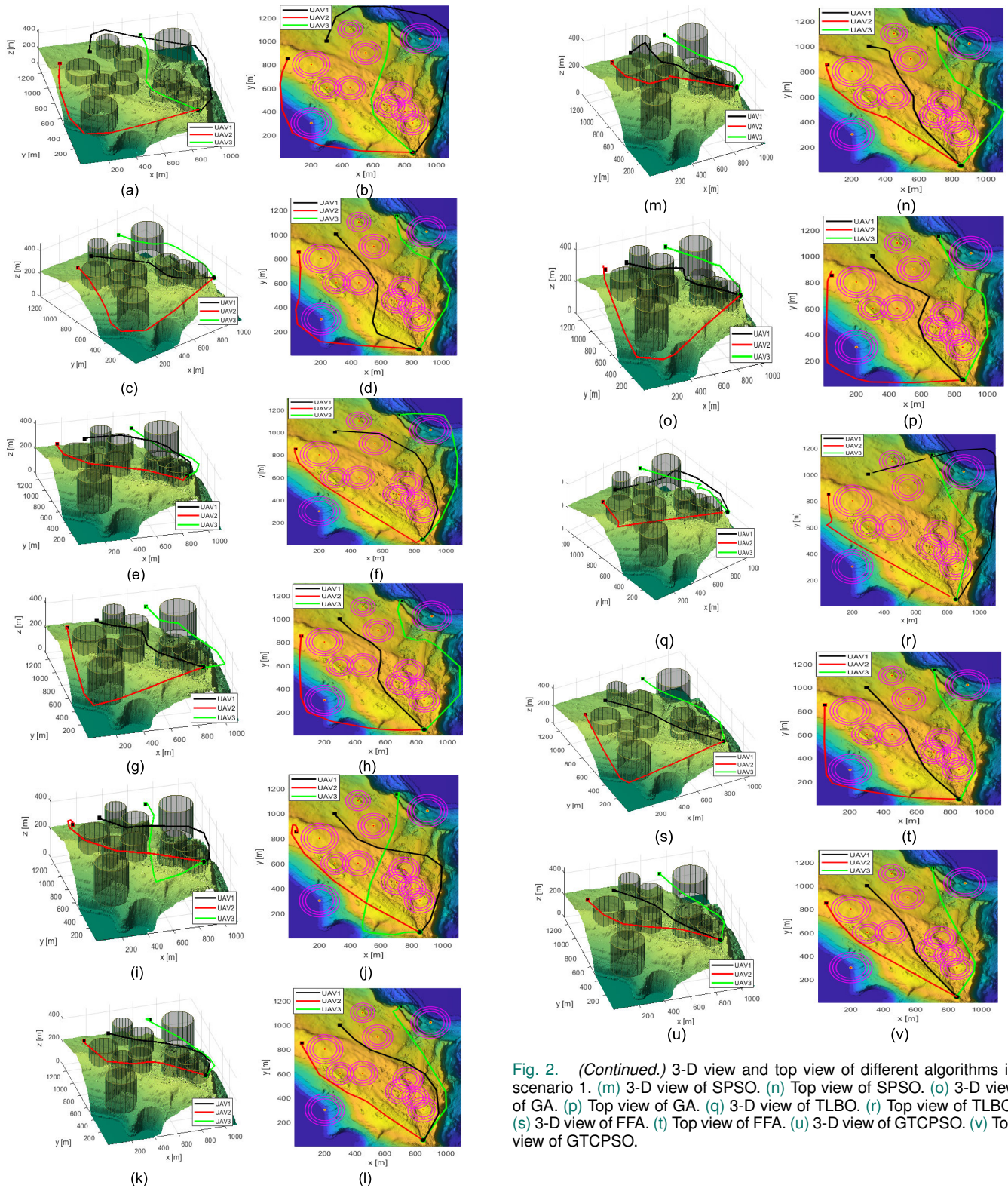


Fig. 2. 3-D view and top view of different algorithms in scenario 1. (a) 3-D view of PSO. (b) Top view of PSO. (c) 3-D view of PSO-LDIW. (d) Top view of PSO-LDIW. (e) 3-D view of θ -PSO. (f) Top view of θ -PSO. (g) 3-D view of APSO. (h) Top view of APSO. (i) 3-D view of AWPSO. (j) Top view of AWPSO. (k) 3-D view of PSOTVAC. (l) Top view of PSOTVAC.

that the GTCPSO algorithm has the lowest average fitness value for UAV formation, which reflects the advantages of

Fig. 2. (Continued.) 3-D view and top view of different algorithms in scenario 1. (m) 3-D view of SPSO. (n) Top view of SPSO. (o) 3-D view of GA. (p) Top view of GA. (q) 3-D view of TLBO. (r) Top view of TLBO. (s) 3-D view of FFA. (t) Top view of FFA. (u) 3-D view of GTCPSO. (v) Top view of GTCPSO.

the proposed GTCPSO algorithm. Besides that, it is observed that from Fig. 7, the average fitness values of various PSOs differ greatly in UAV 1, UAV 3, and UAV 5. Among them, it can be seen that the average fitness value for UAV 2 and UAV 4 are very close with different comparison algorithms. As for UAV 1, UAV 3, and UAV 5, PSO and GTCPSO obtain the maximum and minimum average fitness values, respectively. Fig. 6 displays the convergence curve change

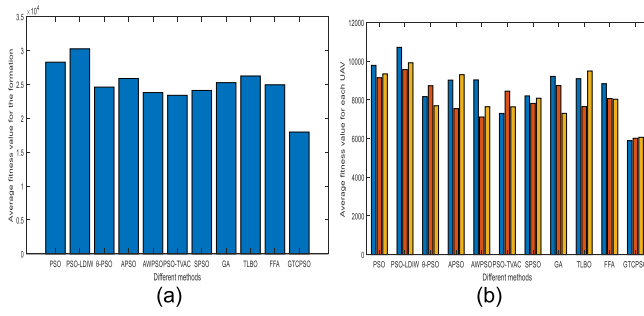


Fig. 3. Fitness value comparison for multiple UAVs and each UAV in scenario 1. (a) Multiple UAVs. (b) Each UAV.

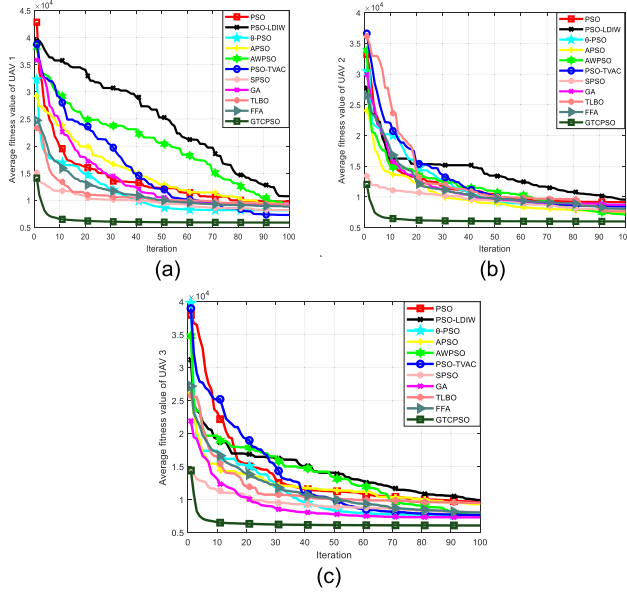


Fig. 4. Convergence curve for each UAV in scenario 2. (a) UAV 1. (b) UAV 2. (c) UAV 3.

of different algorithms for each UAV. It is obvious that the convergence speed of the fitness value of GTCPSP is faster than that of PSO, θ -PSO, APSO, AWPSO, PSO-TVAC, SPSP, GA, TLBO, and FFA.

Compared with other optimization algorithms, such as evolutionary algorithms like GA, TLBO and FFA. For each UAV in scenarios 1 and 2, GTCPSP outperforms other algorithms in both mean and standard. Because FFA introduces the attraction, its attraction, however, decreases with distance when two fireflies are far apart. The brighter firefly will not attract the darker firefly; the entire population can be automatically subdivided into subgroups, each of which can surround each mode or local maximum. FFA performs well in complex scenarios, and therefore, has stable convergence with small deviations. GA and TLBO performed relatively well, but GA was the most volatile. From Figs. 2 and 5, GTCPSP can generate the smoothest and shortest path. GA, TLBO, and FFA do not obtain optimal paths for scenarios 1 and 2. GA searches through crossover and variation to produce high-quality solutions, but it is limited in its search for optimal solutions. Because of the decrease in search dimension, GA converges to the precocious solution quickly. Because of

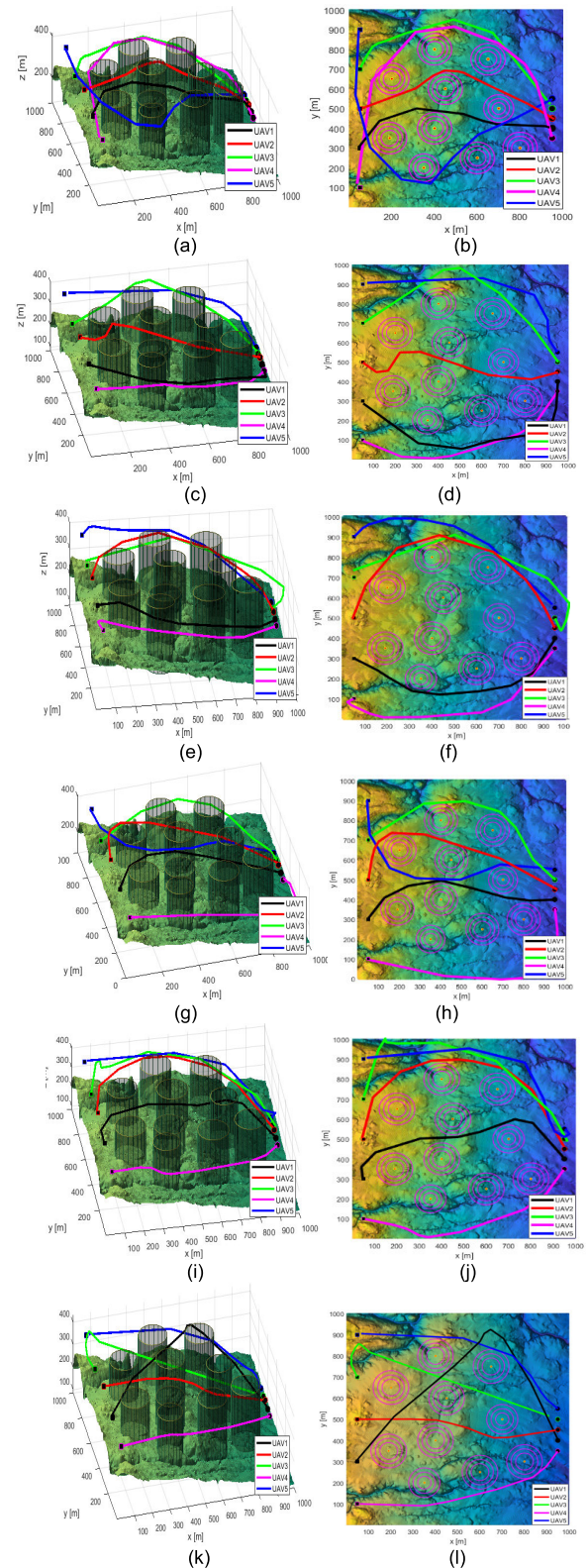


Fig. 5. 3-D view and top view of different algorithms in scenario 2. (a) 3-D view of PSO. (b) Top view of PSO. (c) 3-D view of PSO-LDIW. (d) Top view of PSO-LDIW. (e) 3-D view of θ -PSO. (f) Top view of θ -PSO. (g) 3-D view of APSO. (h) Top view of APSO. (i) 3-D view of AWPSO. (j) Top view of AWPSO. (k) 3-D view of PSOTVAC. (l) Top view of PSOTVAC.

the speed of search through GT technology, GTCPSP has fast enough convergence.

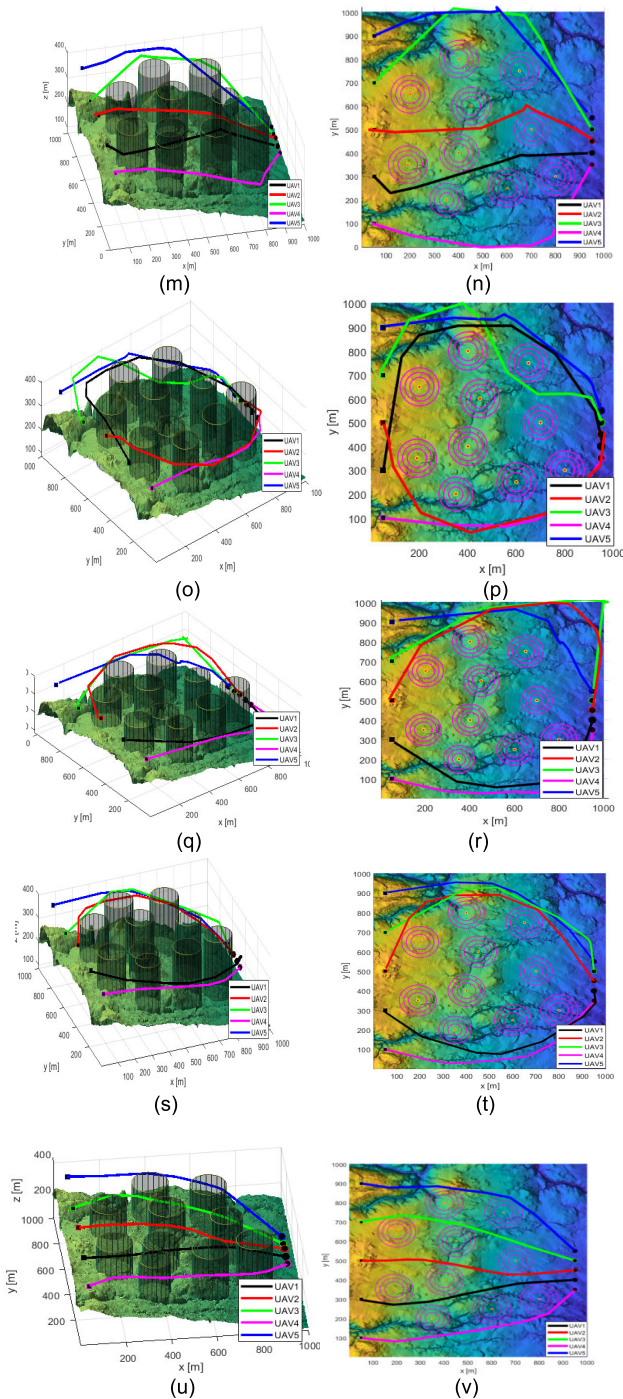


Fig. 5. (Continued.) 3-D view of different algorithms in scenario 2. (m) 3-D view of SPSO. (n) Top view of SPSO. (o) 3-D view of GA. (p) Top view of GA. (q) 3-D view of TLBO. (r) Top view of TLBO. (s) 3-D view of FFA. (t) Top view of FFA. (u) 3-D view of GTCPSO. (v) Top view of GTCPSO.

In the convergence curve, the initial fitness values of SPSO and GTCPSO algorithms obviously outperform other comparison algorithms. This is because spherical coordinates and cylindrical coordinates to represent the waypoints can better meet the dynamic constraints in path planning. In the next search, the GTCPSO with GT technology can seek better quality paths than the SPSO.

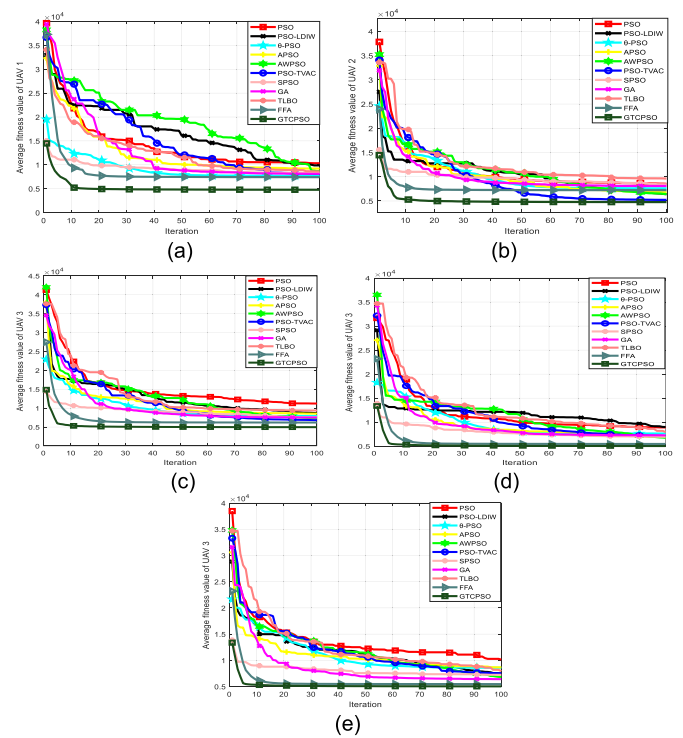


Fig. 6. Convergence curve for each UAV in scenario 2. (a) UAV 1. (b) UAV 2. (c) UAV 3. (d) UAV 4. (e) UAV 5.

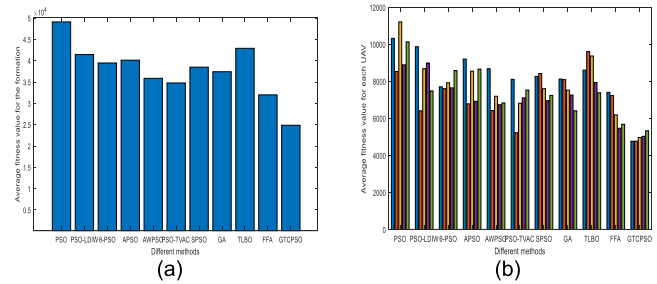


Fig. 7. Fitness value comparison for multiple UAVs and each UAV in scenario 2. (a) Multiple UAVs. (b) Each UAV.

Overall, it indicates GTCPSO has better performance than the other comparative PSO variants in finding safe and feasible routes for multiple UAVs.

V. CONCLUSION AND FUTURE WORK

In order to effectively solve the path planning problem of multiple UAVs in a 3-D complex environment, the GT technology and cylindrical coordinates are incorporated into the traditional PSO algorithm to seek the flyable and safe trajectory for multiple UAVs. First, a multiple UAVs optimization model is established by considering all constraints as the objective function. Then, in view of the shortcomings of the existing PSO algorithm in solving the multiple UAVs' path planning in complex environments, such as low search efficiency and easy to fall into the local optimal solution, gene detection technology is used to find the bottleneck waypoint in the current global best path, and the bottleneck waypoint will be replaced in an appropriate way according to the two

updating strategies, depending on the evolutionary iteration of the population.

The preference of the GTCPSO algorithm proposed in this article is tested by the path planning simulation experiments of single UAVs and multi-UAVs under multiple scenarios. Compared with other existing PSO algorithms, e.g., APSO, AWPSON, PSO-LDIW, θ -PSO, SPSO, PSO-TVAC, GA, TLBO, FFA, the proposed GTCPSO algorithm can generate smoother and more accurate planning routes. The simulation results show that the fitness function value of GTCPSO is better than PSO by 36.4% in complex scenario 1, and is reduced by 28.7%, 31.5%, and 27.9% compared with GA, TLBO, and FFA. Compared with PSO-LDIW, θ -PSO, APSO, AWPSON, PSO-TVAC, and SPSO, the fitness values decrease by 40.5%, 26.9%, 30.5%, 24.4%, 23.1%, and 25.4%, respectively. In complex scenario 2, the fitness function value is optimized by 49.3% compared with PSO and 33.5%, 42.1%, and 22.3% compared with GA, TLBO and FFA. Compared with PSO-LDIW, θ -PSO, APSO, AWPSON, PSO-TVAC, and SPSO, the fitness values decrease by 39.9%, 36.8%, 38.0%, 30.6%, 28.5%, and 35.3%, respectively.

In future work, we will consider large-scale, heterogeneous UAV confrontation and dynamic complex combat scenarios. More attention will be placed on real-time multiple UAVs flight path planning algorithm.

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